# Parametric representations and Cartesian equations

## Lecture and worked examples

Maths Explorer

# Objectives

Describe a line and a plane; determine intersections; choose a method for a projection.

## Course notes

Throughout this chapter, coordinates are given in an orthonormal frame of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to compute distances.

# Parametric lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector $(a, b, c)$ has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \qquad (t \in \mathbb{R}).$$

A point belongs to the line if there exists *one and the same* real number $t$ satisfying the three equalities. A line has many representations: change the starting point or multiply the direction vector by a non-zero real number.

## Parametric lines

$(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

## Cartesian planes

If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector $(a, b, c)$. The plane with normal $\vec{n}$ through $A$ is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

## Cartesian planes

The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, any point satisfying this equation satisfies the orthogonality condition, and therefore belongs to the plane.

## Cartesian planes

The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is $2(x-1) - (y-2) + 3z = 0$, that is $2x - y + 3z = 0$.

## Intersections

For a line and a plane, substitute the parametric expressions for $x, y, z$ into the equation of the plane. A single solution in $t$ gives a single intersection point; an identity gives a line contained in the plane; a contradiction gives a parallel, disjoint line. For two planes, non-collinear normals give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normals are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use independent parameters $s, t$ and solve the three equations. With no solution and non-collinear direction vectors, the lines are non-coplanar.

## Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, find the projection of $M$ on the normal line through $M$: $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine $t$, then compute $MH$. We deduce, without making it a formula to be learned in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

## Projection onto a plane

After computing a projection, check simultaneously that $H \in \mathcal{P}$ and that $\overrightarrow{MH}$ is collinear with the normal vector.

# Research activity

**Three lines facing a plane - 30 min.**

In an orthonormal frame of space, consider the plane $\mathcal{P}$ with equation $x + y + z = 3$. In each question, the parameter $t$ ranges over $\mathbb{R}$.

## Research activity (2)

1. The line $d_1$ has the representation $(x; y; z) = (t; t; t)$. Determine $d_1 \cap \mathcal{P}$.
2. The line $d_2$ has the representation $(x; y; z) = (3 + t; -t; 0)$. Determine $d_2 \cap \mathcal{P}$.
3. The line $d_3$ has the representation $(x; y; z) = (t; -t; 0)$. Determine $d_3 \cap \mathcal{P}$.

## Research activity (3)

**4.** For each line, state whether the equation obtained after substituting into that of the plane has a unique solution, no solution, or every real number as a solution. Relate these cases to the geometric positions found.